

# Gender Differences in Skippable Ad Acceptance Across Ad Brand Categories

*Marketing Models — Assignment 1*

**Group 11**

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## Research Question & Framework

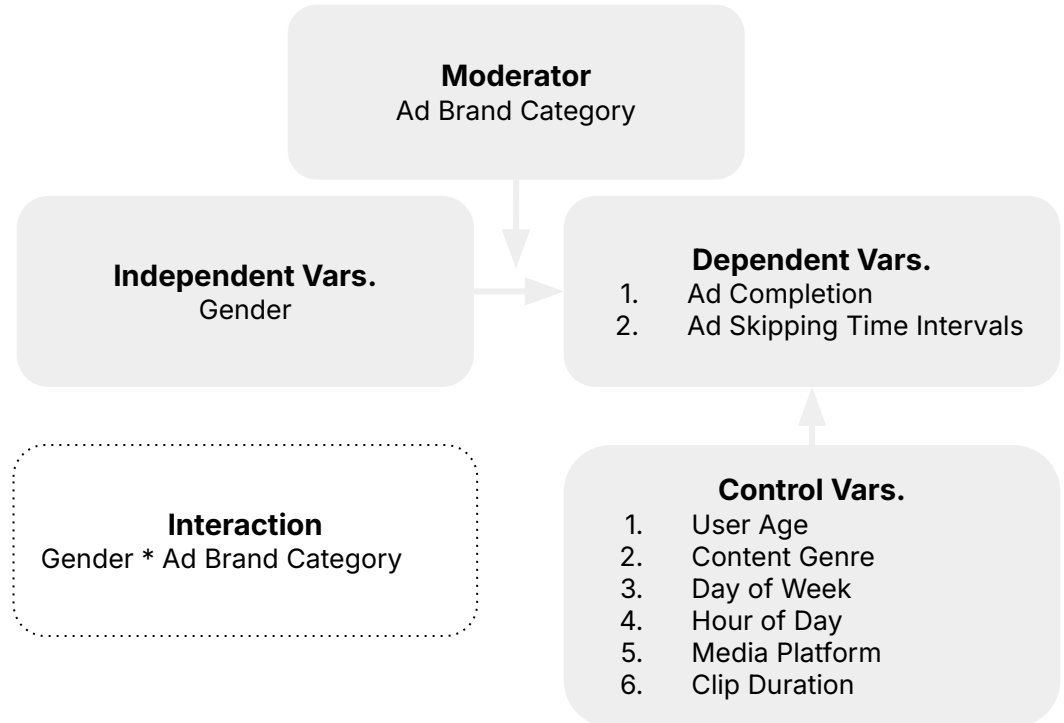
### Business Question:

*Provide useful strategies to improve pre-roll skippable ad acceptance.*

### Research Question:

*How do **men and women** differ in their acceptance of skippable ads **across different brand categories**?*

### Conceptual Framework



## Data Overview

|              |                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| Dataset      | pre-roll_ad.csv                                                                                                                 |
| Period       | July 1–28, 2019                                                                                                                 |
| Observations | 10,000 users who were exposed to 15-second pre-roll skippable video ads on a South Korean popular online video content platform |

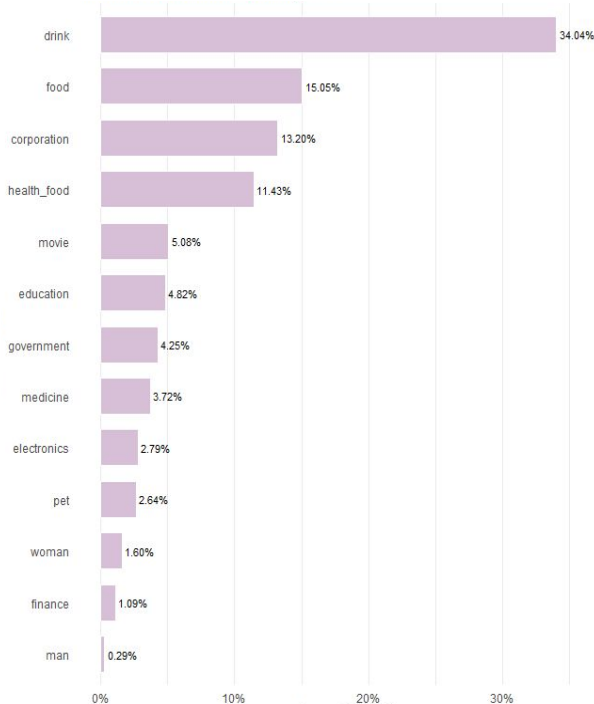
## Variables

| Variable                | Description                                                                                       | Variable       | Description                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------|
| user_id                 | Unique identifier for each user                                                                   | tt             | Hour of the day the ad was viewed                                     |
| age_new                 | User age group measured in 10-year intervals                                                      | media_platform | Platform used to view the ad                                          |
| gender_new              | Gender of the user                                                                                | clip_duration  | The duration of video clips, which follow ads (seconds)               |
| ad_brand_cat            | Ad brand category                                                                                 | ad_complete    | Indicates whether the ad was watched in full                          |
| gender_new:ad_brand_cat | Interaction term to examine how the effect of ad brand category on ad acceptance varies by gender | l_ad_stop      | Lower limit of time interval during which an ad was skipped (seconds) |
| genre_new               | Program genre of the content preceding the ad                                                     | u_ad_stop      | Upper limit of time interval during which an ad was skipped (seconds) |
| day                     | Day of the week the ad was viewed                                                                 |                |                                                                       |

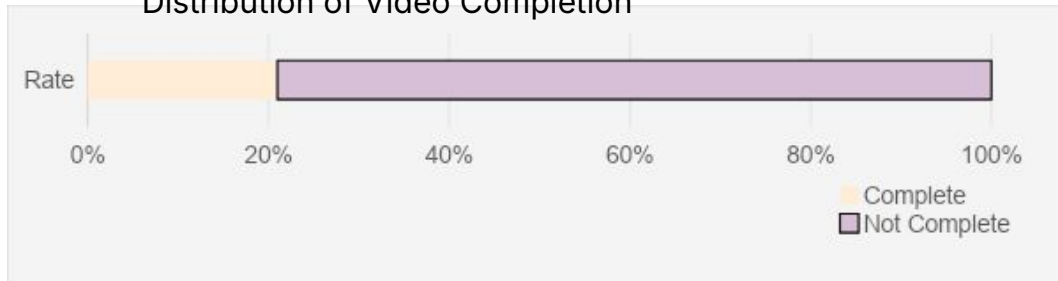
# Descriptive Statistics Analysis

Total observations of **10,000** of the data set on a South Korean popular online video content platform

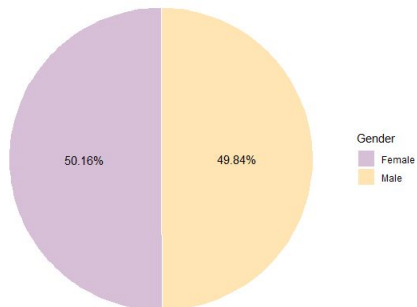
## Distribution of Ad Brand Categories



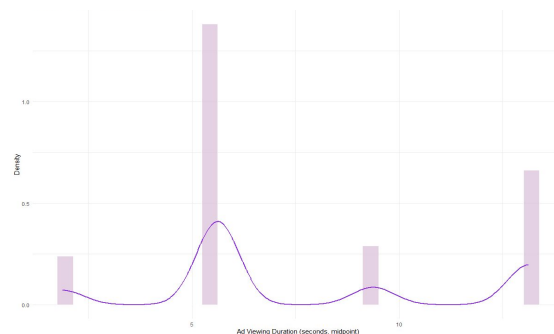
## Distribution of Video Completion



## Distribution of Gender



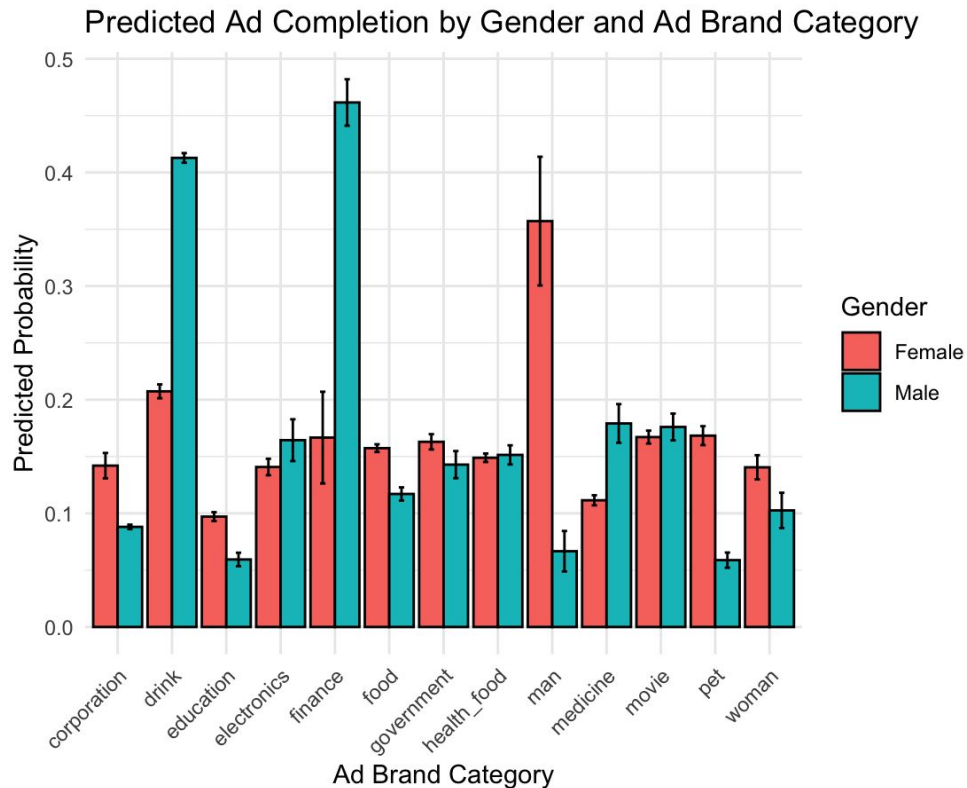
## Distribution of Ad Viewing Durations



Mean:7.62 Median:5.62 Sd:3.65

## Preliminary Analysis: Gender Differences in Ad Completion by Ad Brand Category

Using summarySE()



- **Variations in ad completion** between men and women notable across certain ad categories
  - Men predicted to complete **drink & finance-related ads** significantly more than women
  - Women predicted to complete **male-related ads** significantly more than men
- This graph provides a **strong rationale for analyzing the interaction** between gender and ad brand category

# Output: Logit Regression Model

```
glm(formula = ad_complete ~ gender_new * ad_brand_cat + media_platform + tt + day + age_new +
genre_new + clip_duration, family = binomial(link = logit), data = ad)
```

| Coefficient of Interaction Term (Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )     |
|--------------------------------------------------------------|----------|------------|---------|--------------|
| gender_newMale                                               | -0.869   | 0.256      | -3.389  | 0.0007 ***   |
| Male x Drink                                                 | 1.205    | 0.272      | 4.435   | 9.23e-06 *** |
| Male x Education                                             | 0.180    | 0.526      | 0.342   | 0.732        |
| Male x Electronics                                           | 0.889    | 0.460      | 1.932   | 0.053        |
| Male x Finance                                               | 1.974    | 0.728      | 2.713   | 0.007 **     |
| Male x Food                                                  | 0.197    | 0.311      | 0.632   | 0.528        |
| Male x Government                                            | 0.410    | 0.405      | 1.011   | 0.312        |
| Male x Health Food                                           | 0.629    | 0.316      | 1.986   | 0.047 *      |
| Male x Man                                                   | -1.361   | 1.217      | -1.119  | 0.263        |
| Male x Medicine                                              | 1.323    | 0.453      | 2.921   | 0.003 **     |
| Male x Movie                                                 | 0.804    | 0.376      | 2.140   | 0.032 *      |
| Male x Pet                                                   | -0.756   | 0.610      | -1.238  | 0.216        |
| Male x Woman                                                 | 0.268    | 0.649      | 0.413   | 0.680        |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Null Deviance: 10313.5 on 9999 degrees of freedom  
Residual Deviance: 9129.1 on 9953 degrees of freedom  
AIC: 9223.1

*\*only displaying output of gender  
& interaction term*

## Likelihood Ratio Tests: Overall Model & Interaction Significance

- LRT for overall significance of full model**  
Ho: None of the variables in the model help explain variation in skippable ad acceptance  
**p-value ~0 \*\*\* → Conclusion: Reject  $H_0$ , Full model helps explain ad acceptance better than a model with no predictors**
- LRT for significance on Interaction Term (Gender x Ad Brand)**  
Ho: There's no difference in how men and women accept ads across brand categories  
**p-value ~0 \*\*\* → Conclusion: Reject  $H_0$ , Gender differences in ad acceptance vary significantly by at least one ad brand category**

## Output: Logit Regression Model (Cont.)

```
glm(formula = ad_complete ~ gender_new * ad_brand_cat + media_platform + tt + day + age_new +
genre_new + clip_duration, family = binomial(link = logit), data = ad)
```

| Coefficient of Interaction Term (Gender x Ad Brand Category) | Estimate     | Std. Error   | z value      | Pr(> z )            |
|--------------------------------------------------------------|--------------|--------------|--------------|---------------------|
| gender_newMale                                               | -0.869       | 0.256        | -3.389       | 0.0007 ***          |
| <b>Male x Drink</b>                                          | <b>1.205</b> | <b>0.272</b> | <b>4.435</b> | <b>9.23e-06 ***</b> |
| Male x Education                                             | 0.180        | 0.526        | 0.342        | 0.732               |
| Male x Electronics                                           | 0.889        | 0.460        | 1.932        | 0.053               |
| <b>Male x Finance</b>                                        | <b>1.974</b> | <b>0.728</b> | <b>2.713</b> | <b>0.007 **</b>     |
| Male x Food                                                  | 0.197        | 0.311        | 0.632        | 0.528               |
| Male x Government                                            | 0.410        | 0.405        | 1.011        | 0.312               |
| <b>Male x Health Food</b>                                    | <b>0.629</b> | <b>0.316</b> | <b>1.986</b> | <b>0.047 *</b>      |
| Male x Man                                                   | -1.361       | 1.217        | -1.119       | 0.263               |
| <b>Male x Medicine</b>                                       | <b>1.323</b> | <b>0.453</b> | <b>2.921</b> | <b>0.003 **</b>     |
| <b>Male x Movie</b>                                          | <b>0.804</b> | <b>0.376</b> | <b>2.140</b> | <b>0.032 *</b>      |
| Male x Pet                                                   | -0.756       | 0.610        | -1.238       | 0.216               |
| Male x Woman                                                 | 0.268        | 0.649        | 0.413        | 0.680               |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Null Deviance: 10313.5 on 9999 degrees of freedom  
Residual Deviance: 9129.1 on 9953 degrees of freedom

AIC: 9223.1

*\*only displaying output of gender  
& interaction term*

### Wald Test: Key Interaction Effects

- **Male x Drink:** The odds of ad completion for male viewers watching drink-related ads are **39.9% higher** than the odds of ad completion for female viewers on drink-related ads (statistically significant at 0.1% level,  $p \sim 0$ )
- **Male x Finance:** The odds of ad completion for male viewers watching finance-related ads are **201.9% higher** than the odds of ads completion for female viewers on finance-related ads (statistically significant at 1% level,  $p=0.007$ )
- **Male x Health Food:** The odds of ad completion for male viewers watching health food-related ads are **21.34% lower** than the odds of ads completion for female viewers on health food-related ads (statistically significant at 5% level,  $p=0.047$ )
- **Male x Medicine:** The odds of ad completion for male viewers watching medicine-related ads are **57.46% higher** than the odds of ads completion for female viewers on medicine-related ads (statistically significant at 1% level,  $p=0.003$ )
- **Male x Movie:** The odds of ad completion for male viewers watching movie-related ads are **6.29% lower** than the odds of ads completion for female viewers on movie-related ads (statistically significant at 5% level,  $p=0.032$ )



# Output: Probit Regression Model

```
glm(formula = ad_complete ~ gender_new * ad_brand_cat + media_platform + tt + day + age_new +
genre_new + clip_duration, family = binomial(link = probit), data = ad)
```

| Coefficient of Interaction Term<br>(Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )     |
|-----------------------------------------------------------------|----------|------------|---------|--------------|
| gender_newMale                                                  | -0.489   | 0.138      | -3.551  | 0.0004 ***   |
| Male x Drink                                                    | 0.719    | 0.147      | 4.904   | 9.38e-07 *** |
| Male x Education                                                | 0.108    | 0.264      | 0.410   | 0.681        |
| Male x Electronics                                              | 0.509    | 0.251      | 2.031   | 0.042 *      |
| Male x Finance                                                  | 1.189    | 0.412      | 2.885   | 0.004 **     |
| Male x Food                                                     | 0.152    | 0.166      | 0.917   | 0.359        |
| Male x Government                                               | 0.274    | 0.218      | 1.254   | 0.210        |
| Male x Health Food                                              | 0.372    | 0.171      | 2.181   | 0.029 *      |
| Male x Man                                                      | -0.707   | 0.623      | -1.135  | 0.256        |
| Male x Medicine                                                 | 0.733    | 0.248      | 2.951   | 0.003 **     |
| Male x Movie                                                    | 0.452    | 0.206      | 2.193   | 0.028 *      |
| Male x Pet                                                      | -0.314   | 0.296      | -1.061  | 0.288        |
| Male x Woman                                                    | 0.181    | 0.341      | 0.532   | 0.594        |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Null Deviance: 10313.5 on 9999 degrees of freedom  
Residual Deviance: 9148.7 on 9953 degrees of freedom

AIC: 9242.7

*\*only displaying output of gender  
& interaction term*

## Likelihood Ratio Tests:

### Overall Model & Interaction Significance

#### 1. LRT for overall significance of full model

Ho: None of the variables in the model help explain variation in skippable ad acceptance

**p-value ~0 \*\*\* → Conclusion: Reject  $H_0$ , Full model helps explain ad acceptance better than a model with no predictors**

#### 2. LRT for significance on interaction term

Ho: There's no difference in how men and women accept ads across brand categories

**p-value ~0 \*\*\* → Conclusion: Reject  $H_0$ , Gender differences in ad acceptance vary significantly by at least one ad brand category**

# Output: Probit Regression Model

```
glm(formula = ad_complete ~ gender_new * ad_brand_cat + media_platform + tt + day + age_new +
genre_new + clip_duration, family = binomial(link = probit), data = ad)
```

| Coefficient of Interaction Term<br>(Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )     |
|-----------------------------------------------------------------|----------|------------|---------|--------------|
| gender_newMale                                                  | -0.489   | 0.138      | -3.551  | 0.0004 ***   |
| Male x Drink                                                    | 0.719    | 0.147      | 4.904   | 9.38e-07 *** |
| Male x Education                                                | 0.108    | 0.264      | 0.410   | 0.681        |
| Male x Electronics                                              | 0.509    | 0.251      | 2.031   | 0.042 *      |
| Male x Finance                                                  | 1.189    | 0.412      | 2.885   | 0.004 **     |
| Male x Food                                                     | 0.152    | 0.166      | 0.917   | 0.359        |
| Male x Government                                               | 0.274    | 0.218      | 1.254   | 0.210        |
| Male x Health Food                                              | 0.372    | 0.171      | 2.181   | 0.029 *      |
| Male x Man                                                      | -0.707   | 0.623      | -1.135  | 0.256        |
| Male x Medicine                                                 | 0.733    | 0.248      | 2.951   | 0.003 **     |
| Male x Movie                                                    | 0.452    | 0.206      | 2.193   | 0.028 *      |
| Male x Pet                                                      | -0.314   | 0.296      | -1.061  | 0.288        |
| Male x Woman                                                    | 0.181    | 0.341      | 0.532   | 0.594        |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Null Deviance: 10313.5 on 9999 degrees of freedom  
Residual Deviance: 9148.7 on 9953 degrees of freedom

AIC: 9242.7

*\*only displaying output of gender  
& interaction term*

## Wald Test Results: Key Interaction Effects

- **Male x Drink:** For men watching drink-related ads, the z-score of the probability of ad completion **increases by 0.23** compared to women watching drink-related ads (statistically significant at 0.1% level,  $p \sim 0$ )
- **Male x Electronics:** For men watching electronics-related ads, the z-score of the probability of ad completion **increases by 0.02** compared to women watching electronics-related ads (statistically significant at 5% level,  $p=0.042$ )
- **Male x Finance:** For men watching finance-related ads, the z-score of the probability of ad completion **increases by 0.7** compared to women watching finance-related ads (statistically significant at 1% level,  $p=0.004$ )
- **Male x Health Food:** For men watching health food-related ads, the z-score of the probability of ad completion **decreases by 0.117** compared to women watching health food-related ads (statistically significant at 5% level,  $p=0.029$ )
- **Male x Medicine:** For men watching medicine-related ads, the z-score of the probability of ad completion **increases by 0.244** compared to women watching medicine-related ads (statistically significant at 1% level,  $p=0.003$ )
- **Male x Movie:** For men watching movie-related ads, the z-score of the probability of ad completion **decreases by 0.037** compared to women watching movie-related ads (statistically significant at 5% level,  $p=0.028$ )

# Output: Linear Probability Model

```
lm(formula = ad_complete ~ gender_new * ad_brand_cat + media_platform + tt + day + age_new +  
genre_new + clip_duration, data = ad)
```

| Coefficient of Interaction Term (Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )     |
|--------------------------------------------------------------|----------|------------|---------|--------------|
| gender_newMale                                               | -0.115   | 0.034      | -3.445  | 0.0006 ***   |
| Male x Drink                                                 | 0.203    | 0.036      | 5.664   | 1.52e-08 *** |
| Male x Education                                             | 0.060    | 0.054      | 1.099   | 0.272        |
| Male x Electronics                                           | 0.117    | 0.062      | 1.886   | 0.059        |
| Male x Finance                                               | 0.332    | 0.104      | 3.184   | 0.001 **     |
| Male x Food                                                  | 0.031    | 0.040      | 0.769   | 0.442        |
| Male x Government                                            | 0.051    | 0.053      | 0.960   | 0.337        |
| Male x Health Food                                           | 0.084    | 0.042      | 2.013   | 0.044 *      |
| Male x Man                                                   | -0.191   | 0.147      | -1.302  | 0.193        |
| Male x Medicine                                              | 0.168    | 0.062      | 2.721   | 0.007 **     |
| Male x Movie                                                 | 0.106    | 0.052      | 2.049   | 0.040 *      |
| Male x Pet                                                   | -0.063   | 0.063      | 0.991   | 0.322        |
| Male x Woman                                                 | 0.045    | 0.078      | 0.582   | 0.561        |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Residual Standard Error: 0.3837 on 9953 degrees of freedom

Multiple R-Squared: 0.1207, Adjusted R-Squared: 0.1167

F-statistic: 29.71 on 46 and 9953 DF, p-value: < 2.2e-16

*\*only displaying output of gender  
& interaction term*

## F-Test: Overall Model Significance

- Ho: The model does not explain any variation in the dependent variable
- p-value~0 \*\*\* → Conclusion: Reject  $H_0$ , At least one coefficient significantly explains variation in ad acceptance**

## ANOVA Test: Interaction Term Significance

- Ho: The reduced model (without interaction) fits the data just as well as the full model (with interaction)
- p-value~0 → Conclusion: Reject  $H_0$ , The full model with the interaction term fits the data significantly better**

## Drop1 Test: Interaction Term Significance

|                             | df | Sum of Sq | RSS    | AIC    | F Value | Pr(>F)           |
|-----------------------------|----|-----------|--------|--------|---------|------------------|
| <none>                      |    |           | 1465.3 | -19111 |         |                  |
| gender_new:ad_br<br>and_cat | 12 | 12.4528   | 1477.8 | -19050 | 7.0486  | 6.293e-13<br>*** |

- Dropping the interaction term would lead to a higher RSS and AIC, indicating worse model fit compared to the full model
- P-value~0 → Conclusion: Dropping the interaction term would significantly worsen model fit**

# Output: Linear Probability Model (Cont.)

```
lm(formula = ad_complete ~ gender_new * ad_brand_cat + media_platform + tt + day + age_new +  
genre_new + clip_duration, data = ad)
```

| Coefficient of Interaction Term (Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )     |
|--------------------------------------------------------------|----------|------------|---------|--------------|
| gender_newMale                                               | -0.115   | 0.034      | -3.445  | 0.0006 ***   |
| Male x Drink                                                 | 0.203    | 0.036      | 5.664   | 1.52e-08 *** |
| Male x Education                                             | 0.060    | 0.054      | 1.099   | 0.272        |
| Male x Electronics                                           | 0.117    | 0.062      | 1.886   | 0.059        |
| Male x Finance                                               | 0.332    | 0.104      | 3.184   | 0.001 **     |
| Male x Food                                                  | 0.031    | 0.040      | 0.769   | 0.442        |
| Male x Government                                            | 0.051    | 0.053      | 0.960   | 0.337        |
| Male x Health Food                                           | 0.084    | 0.042      | 2.013   | 0.044 *      |
| Male x Man                                                   | -0.191   | 0.147      | -1.302  | 0.193        |
| Male x Medicine                                              | 0.168    | 0.062      | 2.721   | 0.007 **     |
| Male x Movie                                                 | 0.106    | 0.052      | 2.049   | 0.040 *      |
| Male x Pet                                                   | -0.063   | 0.063      | 0.991   | 0.322        |
| Male x Woman                                                 | 0.045    | 0.078      | 0.582   | 0.561        |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Residual Standard Error: 0.3837 on 9953 degrees of freedom  
Multiple R-Squared: 0.1207, Adjusted R-Squared: 0.1167  
F-statistic: 29.71 on 46 and 9953 DF, p-value: < 2.2e-16

*\*only displaying output of gender  
& interaction term*

## T-Test Results: Key Interaction Effects

- **Male x Drink:** For men watching drink-related ads, the probability of ad acceptance is **8.8 percentage points higher** compared to women watching drink-related ads (statistically significant at the 0.1% level,  $p \sim 0$ )
- **Male x Finance:** For men watching finance-related ads, the probability of ad acceptance is **21.7 percentage points higher** compared to women watching finance-related ads (statistically significant at the 1% level,  $p=0.001$ )
- **Male x Health Food:** For men watching health food-related ads, the probability of ad acceptance is **3.1 percentage points lower** compared to women watching health food-related ads (statistically significant at the 5% level,  $p=0.044$ )
- **Male x Medicine:** For men watching medicine-related ads, the probability of ad acceptance is **5.3 percentage points higher** compared to women watching medicine-related ads (statistically significant at the 1% level,  $p=0.007$ )
- **Male x Movie:** For men watching movie-related ads, the probability of ad acceptance is **0.9 percentage points lower** compared to women watching movie-related ads (statistically significant at the 5% level,  $p=0.040$ )

# Output: Interval Regression Model

```
survreg(formula = Surv(l_ad_stop, u_ad_stop, cens, type = "interval") ~ gender_new * ad_brand_cat +
media_platform + tt + day + clip_duration + age_new + genre_new, data = ad, dist = "gaussian")
```

| Coefficient of Interaction Term<br>(Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )  |
|-----------------------------------------------------------------|----------|------------|---------|-----------|
| gender_newMale                                                  | -0.612   | 0.305      | -2.00   | 0.0452 *  |
| Male x Drink                                                    | 1.145    | 0.328      | 3.490   | 0.000 *** |
| Male x Education                                                | 0.170    | 0.500      | 0.340   | 0.733     |
| Male x Electronics                                              | 1.026    | 0.569      | 1.800   | 0.071     |
| Male x Finance                                                  | 1.826    | 0.955      | 1.910   | 0.056     |
| Male x Food                                                     | 0.076    | 0.364      | 0.210   | 0.835     |
| Male x Government                                               | 0.061    | 0.490      | 0.120   | 0.901     |
| Male x Health Food                                              | 0.657    | 0.381      | 1.730   | 0.084     |
| Male x Man                                                      | -3.495   | 1.343      | -2.600  | 0.009 **  |
| Male x Medicine                                                 | 1.242    | 0.566      | 2.200   | 0.028 *   |
| Male x Movie                                                    | 0.879    | 0.473      | 1.860   | 0.063     |
| Male x Pet                                                      | -0.162   | 0.580      | -0.280  | 0.780     |
| Male x Woman                                                    | 0.310    | 0.716      | 0.430   | 0.665     |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Gaussian distribution  
Loglik(model) = -13563.9 Loglik(intercept only) = -13930.9  
Chisq = 734.09 on 46 degrees of freedom, p = 9.9e-125

*\*only displaying output of gender  
& interaction term*

## Likelihood Ratio Tests:

### Overall Model & Interaction Significance

1. LRT for overall significance of full model  
Ho: None of the variables in the model help explain variation in skippable ad acceptance  
**p-value ~0 \*\*\* → Conclusion: Reject  $H_0$ , full model helps explain ad acceptance better than a model with no predictors.**
2. LRT for significance on interaction term  
Ho: There's no difference in the time of men and women skip ads across all brand categories  
**p-value ~0 \*\*\* → Conclusion: Reject  $H_0$ , gender differences vary by at least one brand.**

## Output: Interval Regression Model

```
survreg(formula = Surv(l_ad_stop, u_ad_stop, cens, type = "interval") ~ gender_new * ad_brand_cat +
media_platform + tt + day + clip_duration + age_new + genre_new, data = ad, dist = "gaussian")
```

| Coefficient of Interaction Term<br>(Gender x Ad Brand Category) | Estimate | Std. Error | z value | Pr(> z )  |
|-----------------------------------------------------------------|----------|------------|---------|-----------|
| gender_newMale                                                  | -0.612   | 0.305      | -2.00   | 0.0452 *  |
| Male x Drink                                                    | 1.145    | 0.328      | 3.490   | 0.000 *** |
| Male x Education                                                | 0.170    | 0.500      | 0.340   | 0.733     |
| Male x Electronics                                              | 1.026    | 0.569      | 1.800   | 0.071     |
| Male x Finance                                                  | 1.826    | 0.955      | 1.910   | 0.056     |
| Male x Food                                                     | 0.076    | 0.364      | 0.210   | 0.835     |
| Male x Government                                               | 0.061    | 0.490      | 0.120   | 0.901     |
| Male x Health Food                                              | 0.657    | 0.381      | 1.730   | 0.084     |
| Male x Man                                                      | -3.495   | 1.343      | -2.600  | 0.009 **  |
| Male x Medicine                                                 | 1.242    | 0.566      | 2.200   | 0.028 *   |
| Male x Movie                                                    | 0.879    | 0.473      | 1.860   | 0.063     |
| Male x Pet                                                      | -0.162   | 0.580      | -0.280  | 0.780     |
| Male x Woman                                                    | 0.310    | 0.716      | 0.430   | 0.665     |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Gaussian distribution

Loglik(model)= -13563.9 Loglik(intercept only)= -13930.9

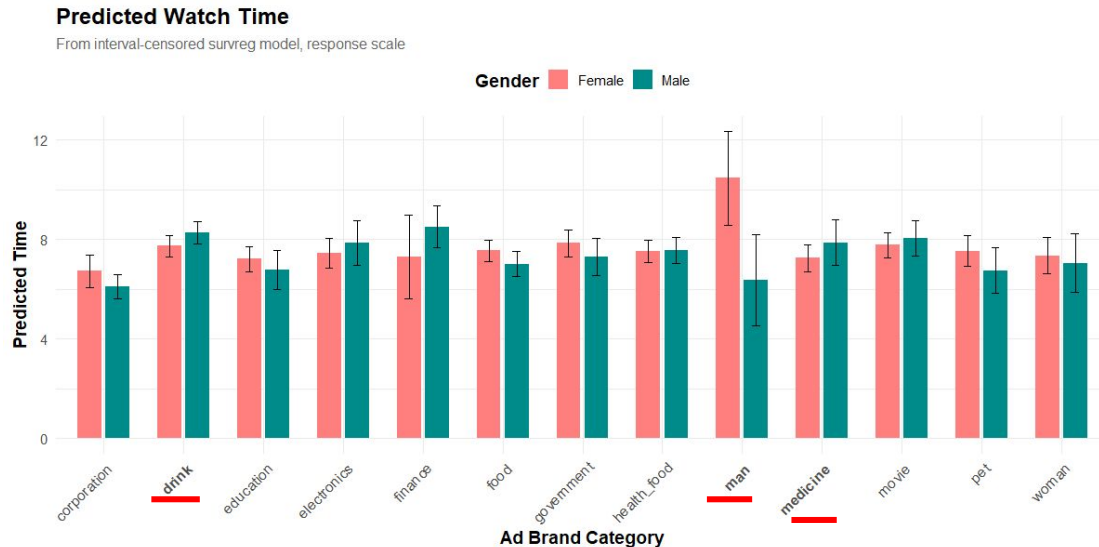
Chisq= 734.09 on 46 degrees of freedom, p= 9.9e-125

### Wald Test Results: Key Interaction Effects

- Male x Drink:** The predicted skip time for male viewers of drink-related ads is **0.53 seconds later** than that for female viewers on drink-related ads (statistically significant at the 0.1% level,  $p = 0.000$ ).
- Male x Man:** The predicted skip time for male viewers of man-related ads is **4.11 seconds earlier** than female viewers on man-related ads (statistically significant at the 1% level,  $p = 0.009$ ).
- Male x Medicine:** The predicted skip time for male viewers of medicine-related ads is **0.63 seconds later** than that for female viewers on medicine-related ads (statistically significant at the 5% level,  $p = 0.028$ ).

# Predicted Seconds Watched — Gender × Brand (model-based)

Holding other variables at typical levels(emmeans)



- Categories where error bars overlap heavily (e.g., movie, health\_food) show no reliable gender gap.
- The differences between men and women are most notable in three categories(drink, man, medicine).
- Although men appeared to stay 1 second longer on financial ads, the sample variability was large (female SE = 0.85 s), the confidence intervals overlapped, and the difference was only marginally significant at the 10% level.

# Output: Weibull Distribution Model

survreg(formula = Surv(Lad\_stop + 0.01, u\_ad\_stop + 0.01, cens, type = "interval") ~ gender\_new \* ad\_brand\_cat + media\_platform + tt + day + clip\_duration + age\_new + genre\_new, data = ad, dist = "weibull")

| Coefficient of Interaction Term<br>(Gender x Ad Brand Category) | Estimate  | Std. Error | z value | Pr(> z )  |
|-----------------------------------------------------------------|-----------|------------|---------|-----------|
| Male x Drink                                                    | 1.51e-01  | 4.03e-02   | 3.76    | 0.000 *** |
| Male x Education                                                | -8.01e-03 | 6.24e-02   | -0.13   | 0.89785   |
| Male x Electronics                                              | 1.28e-01  | 7.02e-02   | 1.83    | 0.06742   |
| Male x Finance                                                  | 1.99e-01  | 1.17e-01   | 1.70    | 0.08884   |
| Male x Food                                                     | 1.64e-02  | 4.52e-02   | 0.36    | 0.71751   |
| Male x Government                                               | 5.47e-02  | 6.06e-02   | 0.90    | 0.36600   |
| Male x Health Food                                              | 9.55e-02  | 4.73e-02   | 0.730   | 0.084     |
| Male x Man                                                      | -2.78e-01 | 1.64e-01   | -1.69   | 0.091 *   |
| Male x Medicine                                                 | 1.67e-01  | 6.97e-02   | 2.39    | 0.01670 * |
| Male x Movie                                                    | 1.22e-01  | 5.85e-02   | 2.08    | 0.03723 * |
| Male x Pet                                                      | 5.13e-03  | 7.19e-02   | 0.07    | 0.94313   |
| Male x Woman                                                    | 6.05e-02  | 8.83e-02   | 0.68    | 0.49344   |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Scale = 0.409  
Weibull distribution  
Loglik(model)= -13195.4    Loglik(intercept only)= -13573.9  
Chisq= 756.96 on 46 degrees of freedom, p= 2.1e-129  
Number of Newton-Raphson Iterations: 4  
n= 10000

# Output: Log-logistic Distribution Model

survreg(formula = Surv(Lad\_stop + 0.01, u\_ad\_stop + 0.01, cens, type = "interval") ~ gender\_new \* ad\_brand\_cat + media\_platform + tt + day + clip\_duration + age\_new + genre\_new, data = ad, dist = "loglogistic")

| Coefficient of Interaction Term<br>(Gender x Ad Brand Category) | Estimate  | Std. Error | z value | Pr(> z )   |
|-----------------------------------------------------------------|-----------|------------|---------|------------|
| Male x Drink                                                    | 1.57e-01  | 4.83e-02   | 3.24    | 0.00119 ** |
| Male x Education                                                | 1.63e-02  | 7.13e-02   | 0.23    | 0.81880    |
| Male x Electronics                                              | 1.31e-01  | 8.12e-02   | 1.61    | 0.10698    |
| Male x Finance                                                  | 2.78e-01  | 1.44e-01   | 1.93    | 0.05363    |
| Male x Food                                                     | 3.48e-05  | 5.29e-02   | 0.00    | 0.99947    |
| Male x Government                                               | -1.13e-02 | 7.25e-02   | -0.16   | 0.87586    |
| Male x Health Food                                              | 8.56e-02  | 5.55e-02   | 1.54    | 0.12287    |
| Male x Man                                                      | -5.56e-01 | 1.94e-01   | -2.86   | 0.00424 ** |
| Male x Medicine                                                 | 1.92e-01  | 8.29e-02   | 2.32    | 0.02023 *  |
| Male x Movie                                                    | 1.20e-01  | 6.87e-02   | 1.74    | 0.08149    |
| Male x Pet                                                      | -3.57e-02 | 8.35e-02   | -0.43   | 0.66857    |
| Male x Woman                                                    | 3.09e-02  | 1.05e-01   | 0.30    | 0.76781    |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

Scale = 0.409  
Log logistic distribution  
Loglik(model)= -13429.1    Loglik(intercept only)= -13807.8  
Chisq= 757.36 on 46 degrees of freedom, p= 1.7e-129  
Number of Newton-Raphson Iterations: 4  
n= 10000



# Summary of Results

Ernest

Binary DV: Ad Completion

Interval-Censored DV: Ad Skipping Time

| Estimated Coefficients                           |                    | Linear Probability Model | Logit Regression Model | Probit Regression Model | Interval Regression Model | Weibull Distribution Model | Log-logistic Distribution Model |
|--------------------------------------------------|--------------------|--------------------------|------------------------|-------------------------|---------------------------|----------------------------|---------------------------------|
| Interaction Terms:<br>Gender x Ad Brand Category | Male x Drink       | 0.203 ***                | 1.205 ***              | 0.719 ***               | 1.145 ***                 | 1.51e-01 ***               | 1.57e-01 **                     |
|                                                  | Male x Education   | 0.060                    | 0.180                  | 0.108                   | 0.170                     | -8.01e-03                  | 1.63e-02                        |
|                                                  | Male x Electronics | 0.117                    | 0.889                  | 0.509 *                 | 1.026                     | 1.28e-01                   | 1.31e-01                        |
|                                                  | Male x Finance     | 0.332 **                 | 1.974 **               | 1.189 **                | 1.826                     | 1.99e-01                   | 2.78e-01                        |
|                                                  | Male x Food        | 0.031                    | 0.197                  | 0.152                   | 0.076                     | 1.64e-02                   | 3.48e-05                        |
|                                                  | Male x Government  | 0.051                    | 0.410                  | 0.274                   | 0.061                     | 5.47e-02                   | -1.13e-02                       |
|                                                  | Male x Health Food | 0.084 *                  | 0.629 *                | 0.372 *                 | 0.657                     | 9.55e-02                   | 8.56e-02                        |
|                                                  | Male x Man         | -0.191                   | -1.361                 | -0.707                  | -3.495 **                 | -2.78e-01 *                | 1.92e-01 **                     |
|                                                  | Male x Medicine    | 0.168 **                 | 1.323 **               | 0.733 **                | 1.242 *                   | 1.67e-01 *                 | 1.92e-01 *                      |
|                                                  | Male x Movie       | 0.106 *                  | 0.804 *                | 0.452 *                 | 0.879                     | 1.22e-01 *                 | 1.22e-01                        |
|                                                  | Male x Pet         | -0.063                   | -0.756                 | -0.314                  | -0.162                    | 5.13e-03                   | -3.57e-02                       |
|                                                  | Male x Woman       | 0.045                    | 0.268                  | 0.181                   | 0.310                     | 6.05e-02                   | 3.09e-02                        |

\*\*\* significant at 0.1% level; \*\* significant at 1% level, \* significant at 5% level

# Conclusion

## *Answering the Research Question:*

***RQ: How do men and women differ in their acceptance of skippable ads across different brand categories?***

- In terms of **ad completion**:
  - **Male viewers are significantly more likely to complete drink, finance, and medicine-related ads** than their female counterparts
  - **Female viewers are significantly more likely to complete health food and movie-related ads** than their male counterparts
- In terms of **ad skip time**:
  - **Male viewers tend to skip drink and medicine-related ads significantly later** than their female counterparts
  - **Female viewers tend to skip man-related ads significantly later** than their male counterparts

## *Answering the Business Question:*

***BQ: Provide useful strategies to improve pre-roll skippable ad acceptance.***

- Use platform data to identify audience distribution by gender, then align ad categories with the preferences identified in our research, targeting the most relevant audience segments
- For ad categories with low completion rates, consider using non-skippable ads to boost engagement

## Hypothesis of the Results: Link to Social Norms

### *Female Audience*

- Common for women to have health-related goals, explaining **higher completion rates for health food ads than men**
- **Later skip time for male-related ads** may be due to curiosity, unfamiliarity with male products, or shopping for male family members/partners

### *Male Audience*

- Often encouraged to be financially successful and independent, which explains why they are **significantly more receptive to finance-related ads**
- **Higher ad acceptance for drink ads** may be due to:
  - Physique/bodybuilding goals, leading to engagement with protein drink ads
  - Interest in energy drink ads for focus
  - Alcohol ads for socializing and stress relief

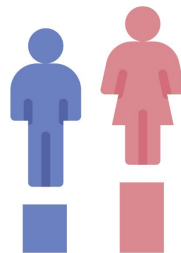
# Marketing Implications

## Ad Placement by Audience Demographics

- **Be strategic in choosing video genres and content creators for ad placements based on gender distribution**
  - **For male viewer-dominated video content: Focus on drink, finance, and medicine-related ads**
  - **For female viewer-dominated video content: Focus on health food, movie, and male-related ads**
- Evaluate the male-to-female audience ratio of the video before deciding on ad placement

## Non-Skippable Ads for Low Completion Categories

- **Consider non-skippable ads for low completion categories**, such as health food and movies for men, and medicine, finance, and electronics for women
- Investigate why engagement is low and adjust strategies accordingly (e.g., test different ad content to enhance engagement)



## Limitations

Ad brand category data size varies significantly, especially with fewer male-category ads

Reasons behind observed gender differences in ad acceptance are unclear

Analysis focuses only on ad completion and skip time, not on downstream outcomes

Lack of clarity on the specific content included in each brand category

## Future Research

Expand data collection to balance sample sizes across all brand categories, especially for male-targeted ads

Conduct qualitative research (e.g., interviews, focus groups) to understand motivations behind viewer behavior

Track additional metrics like brand recall, purchase intent, and long-term engagement after ad exposure

Refine and standardize brand category definitions by analyzing the actual ad content characteristics

## Limitations

Study is based on a single video platform, limiting generalizability



Gender categorization is too broad and doesn't capture within-gender preference differences



Creative elements of the ads were not considered in the analysis



Study focused only on one cultural context, limiting global applicability



## Future Research

Replicate the study across different platforms (e.g., YouTube, TikTok) to test if patterns hold

Incorporate finer audience segmentation (e.g., personality traits, interest profiles) to better understand ad engagement

Explore how factors like ad tone, storytelling style, and visuals interact with gender to influence ad acceptance

Examine cross-cultural differences to see how gender-targeted ad strategies perform internationally

# Thank You!

*Questions or Feedback?*

# Appendix: R Codes

## **#Read in dataset**

```
ad = read.csv("pre-roll_ad.csv") %>%
  mutate(
    gender_new = factor(gender_new, levels = c("Female", "Male")),
    ad_brand_cat = factor(ad_brand_cat),
    media_platform = factor(media_platform,
      levels = c("MOBILE APP", "MOBILE WEB", "PC WEB")),
    age_new = factor(age_new, levels = c(10, 20, 30, 40, 50, 60),
      labels = c("10s", "20s", "30s", "40s", "50s", "60s+"))
  )
```

## **#Descriptive Statistics**

```
descriptives <- ad %>%
  summarise(
    mean = mean(mid_stop, na.rm = TRUE),
    median = median(mid_stop, na.rm = TRUE),
    sd = sd(mid_stop, na.rm = TRUE),
    min = min(mid_stop, na.rm = TRUE),
    q1 = quantile(mid_stop, 0.25, na.rm = TRUE),
    q3 = quantile(mid_stop, 0.75, na.rm = TRUE),
    max = max(mid_stop, na.rm = TRUE)
  )

gender_dist <- ad %>%
  count(gender_new) %>%
  mutate(
    prop = n / sum(n)
  )

brand_dist <- ad %>%
  count(ad_brand_cat) %>%
  mutate(
    prop = n / sum(n)
  )
```

## **#Preliminary Interaction Term Analysis (Graph)**

```
library(ggplot2)
library(dplyr)
library(Rmisc)

plot_data <- summarySE(
  data = ad,
  measurevar = "predicted",
  groupvars = c("gender_new", "ad_brand_cat")
)

ggplot(plot_data, aes(x = ad_brand_cat, y = predicted, fill =
  gender_new)) +
  geom_col(position = position_dodge(width = 0.9), color = "black") +
  geom_errorbar(aes(ymin = predicted - ci, ymax = predicted + ci),
    position = position_dodge(width = 0.9), width = 0.2) +
  labs(
    title = "Predicted Ad Completion by Gender and Ad Brand
    Category",
    x = "Ad Brand Category",
    y = "Predicted Probability",
    fill = "Gender"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



# Appendix: R Codes

## **## Logit Regression Model**

### **## Estimation**

```
library(lmtest)
fit_logit = glm(ad_complete ~ gender_new*ad_brand_cat + media_platform + tt + day
+ age_new + genre_new + clip_duration,
               family = binomial(link = logit), data=ad)
```

```
summary(fit_logit)
```

### **## Log-likelihood value of the full model**

```
logLik(fit_logit)
```

### **## Likelihood ratio test (LRT) for overall significance of the full model**

```
library(lmtest)
```

```
lrtest(fit_logit)
```

### **## LRT for testing gender\*ad brand interaction term**

```
logit_no_interact = glm(ad_complete ~ gender_new + ad_brand_cat + media_platform
+ tt + day + age_new + genre_new + clip_duration,
                      family = binomial(link = logit), data = ad)
```

```
lrtest(logit_no_interact, fit_logit)
```

### **## LRT for testing gender main IV**

```
logit_no_gender = glm(ad_complete ~ ad_brand_cat + media_platform + tt + day +
age_new + genre_new + clip_duration,
                    family = binomial(link = "logit"), data = ad)
```

```
lrtest(logit_no_gender, fit_logit)
```

### **## LRT for testing ad brand category main IV**

```
logit_no_adbrand = glm(ad_complete ~ gender_new + media_platform + tt + day +
age_new + genre_new + clip_duration,
                      family = binomial(link = "logit"), data = ad)
```

```
lrtest(logit_no_adbrand, fit_logit)
```

## **## Probit Regression Model**

### **## Estimation**

```
fit_probit = glm(ad_complete ~ gender_new*ad_brand_cat + media_platform + tt +
day + age_new + genre_new + clip_duration,
                family = binomial(link = probit), data=ad)
summary(fit_probit)
```

### **## Log-likelihood value of the full model**

```
logLik(fit_probit)
```

### **## Likelihood ratio test (LRT) for overall significance of the full model**

```
lrtest(fit_probit)
```

### **## LRT for testing gender\*ad brand interaction term**

```
probit_no_interact = glm(ad_complete ~ gender_new + ad_brand_cat +
media_platform + tt + day + age_new + genre_new + clip_duration,
                        family = binomial(link = probit), data = ad)
```

```
lrtest(probit_no_interact, fit_probit)
```

### **## LRT for testing gender\*ad brand interaction term**

```
probit_no_interact = glm(ad_complete ~ gender_new + ad_brand_cat +
media_platform + tt + day + age_new + genre_new + clip_duration,
                        family = binomial(link = probit), data = ad)
```

```
lrtest(probit_no_interact, fit_probit)
```

### **## LRT for testing gender main IV**

```
probit_no_gender = glm(ad_complete ~ ad_brand_cat + media_platform + tt + day +
age_new + genre_new + clip_duration,
                    family = binomial(link = "probit"), data = ad)
```

```
lrtest(probit_no_gender, fit_probit)
```

### **## LRT for testing ad brand category main IV**

```
probit_no_adbrand = glm(ad_complete ~ gender_new + media_platform + tt + day +
age_new + genre_new + clip_duration,
                      family = binomial(link = "probit"), data = ad)
```

```
lrtest(probit_no_adbrand, fit_probit)
```

# Appendix: R Codes

## ## Linear Probability Model

```
fit_LPM = lm(ad_complete ~ gender_new*ad_brand_cat + media_platform + tt + day + age_new +  
genre_new + clip_duration, data=ad)  
summary(fit_LPM)
```

## ## ANOVA for testing gender\*ad brand interaction term

```
LPM_no_interact = lm(ad_complete ~ gender_new + ad_brand_cat + media_platform + tt + day +  
age_new + genre_new + clip_duration, data = ad)
```

```
anova(LPM_no_interact, fit_LPM)
```

## ## DROP1

```
drop1(fit_LPM, test = "F")
```

## ## Interval Regression Model

```
library(survival)  
ad$cens = 3
```

## ## Estimation

```
irm = survreg(Surv(l_ad_stop, u_ad_stop, cens, type="interval") ~ gender_new*ad_brand_cat +  
media_platform + tt + day + clip_duration + age_new + genre_new,  
data=ad, dist="gaussian")  
summary(irm)
```

## ## Likelihood ratio test (LRT) for overall significance of the full model

```
lrtest(irm)
```

## ## Weibull Distribution Model

```
mod_wb <- survreg(  
Surv(l_ad_stop + 0.01, u_ad_stop + 0.01, cens, type="interval") ~  
gender_new * ad_brand_cat + media_platform + tt + day +  
clip_duration + age_new + genre_new,  
data = ad,  
dist = "weibull"  
)
```

## ## Log-logistic Distribution Model

```
mod_loglogistic <- survreg(  
Surv(l_ad_stop + 0.01, u_ad_stop + 0.01, cens, type="interval") ~  
gender_new * ad_brand_cat + media_platform + tt + day +  
clip_duration + age_new + genre_new,  
data = ad,  
dist = "loglogistic"  
)
```

## ## Graph

```
library(ggplot2)  
library(ggtext)
```

```
emm_int <- emmeans(  
object = mod_full,  
specs = ~ gender_new * ad_brand_cat,  
type = "response"  
)
```

```
palette <- c(Female = "#FF7F7F", Male = "#008B8B")  
bold_these <- c("drink", "medicine", "man")
```

```
df_emm <- as.data.frame(emm_int)
```

```
p <- ggplot(df_emm, aes(x = ad_brand_cat, y = emmean, fill = gender_new)) +  
geom_col(position = position_dodge(0.7), width = 0.6) +  
geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL),  
position = position_dodge(0.7), width = 0.2, size = 0.7) +  
scale_fill_manual(values = palette) +  
scale_x_discrete(labels = function(x) {  
  supply(x, function(lab) {  
    if (lab %in% bold_these) paste0("***", lab, "***") else lab  
  })  
}) +  
theme_minimal(base_size = 14, base_family = "Arial") +  
theme(  
axis.text.x = element_markdown(size = 12, angle = 45, hjust = 1),  
axis.title = element_text(face = "bold", size = 14),  
plot.title = element_text(face = "bold", size = 18),  
plot.subtitle = element_text(size = 12, color = "gray40"),  
legend.position = "top",  
legend.title = element_text(face = "bold")  
) +  
labs(  
title = "Predicted Watch Time",  
subtitle = "Interval-censored survreg model (response scale)",  
x = "Ad Brand Category",  
y = "Predicted Seconds Watched",  
fill = "Gender"  
)
```